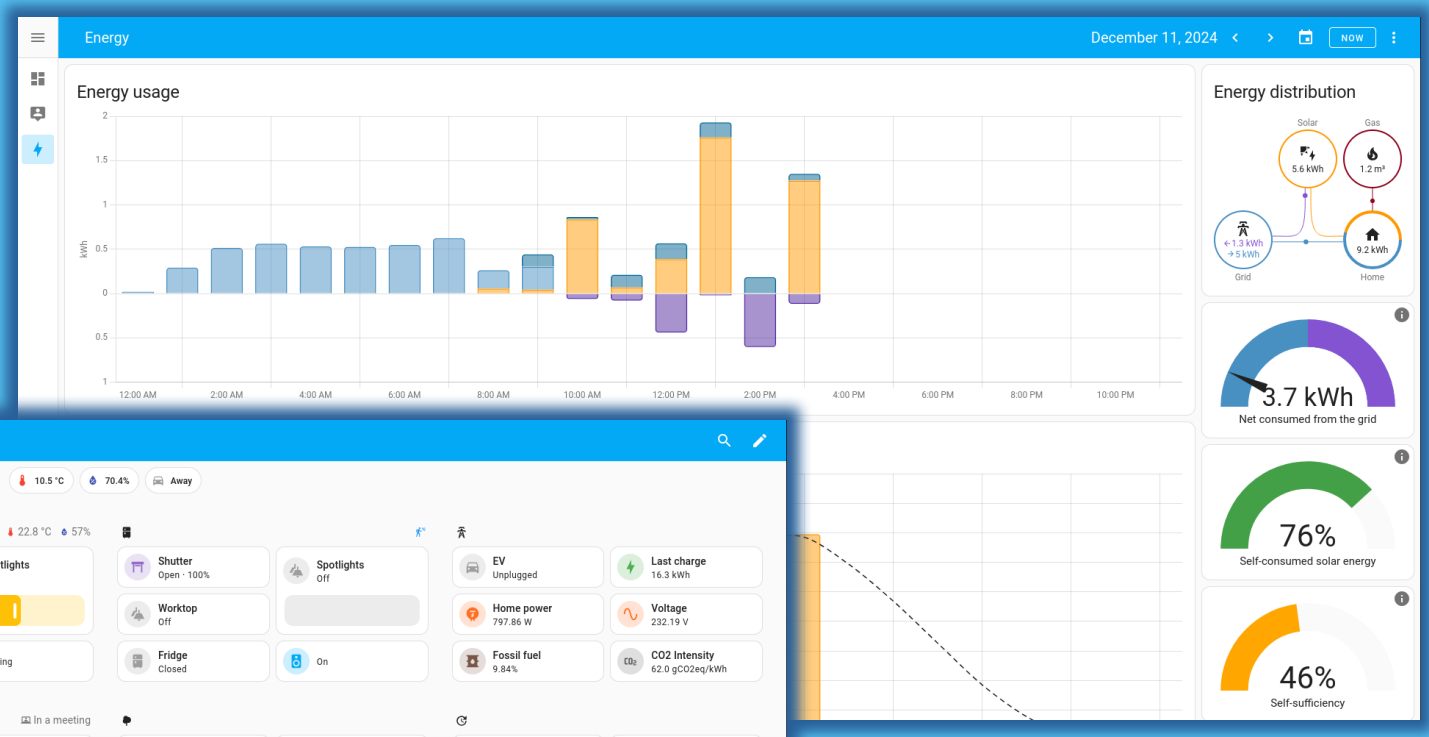
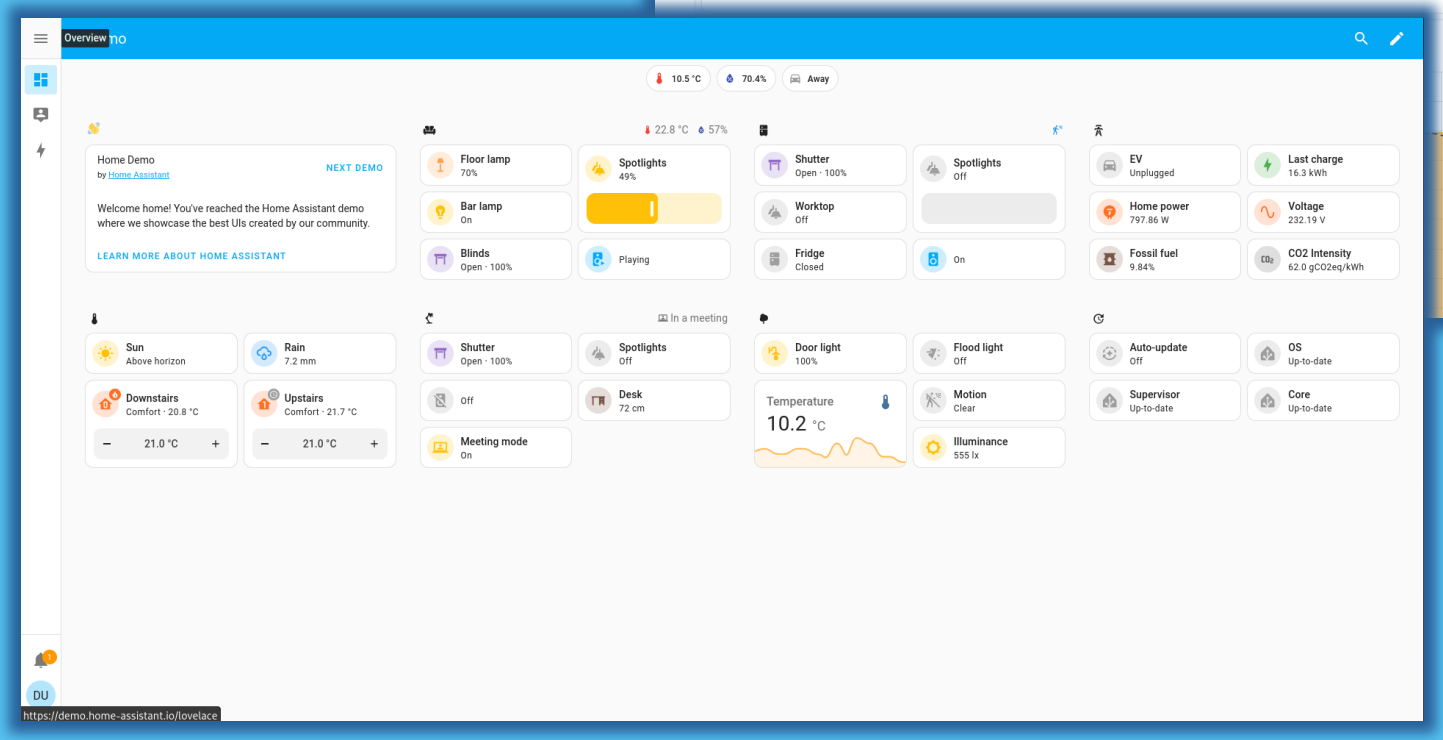
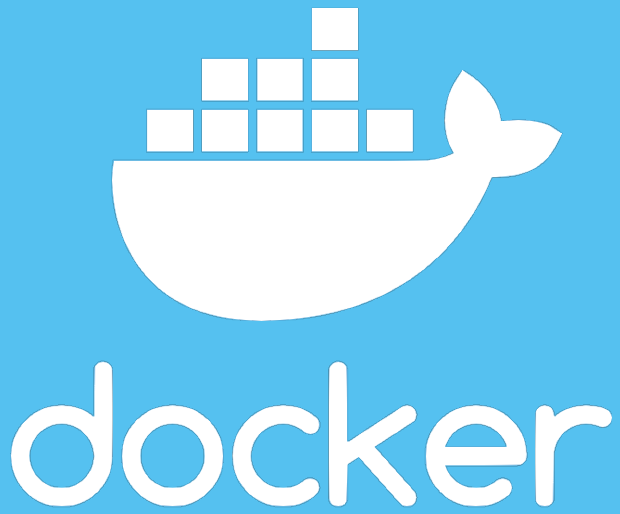
Home Assistant

What?


Energy
December 11, 2024
NOW
Energy usage
kWh
12:00 AM
2:00 AM
4:00 AM
6:00 AM
8:00 AM
10:00 AM
12:00 PM
2:00 PM
4:00 PM
6:00 PM
8:00 PM
10:00 PM
Energy distribution
Solar
Gas
5.6 kWh
1.2 m³
← 1.3 kWh
→ 5 kWh
9.2 kWh
Grid
Home
3.7 kWh
Net consumed from the grid
76%
Self-consumed solar energy
46%
Self-sufficiency


Overview
10.5 °C
70.4%
Away
Home Demo
by Home Assistant
NEXT DEMO
Welcome home! You've reached the Home Assistant demo where we showcase the best UIs created by our community.
LEARN MORE ABOUT HOME ASSISTANT
22.8 °C
57%
Floor lamp
70%
Spotlights
49%
Bar lamp
On
Blinds
Open · 100%
Playing
Shutter
Open · 100%
Spotlights
Off
Worktop
Off
Fridge
Closed
On
EV
Unplugged
Last charge
16.3 kWh
Home power
797.86 W
Voltage
232.19 V
Fossil fuel
9.84%
CO2 Intensity
62.0 gCO2eq/kWh
Sun
Above horizon
Rain
7.2 mm
Downstairs
Comfort · 20.8 °C
21.0 °C
Upstairs
Comfort · 21.7 °C
21.0 °C
In a meeting
Shutter
Open · 100%
Spotlights
Off
Off
Desk
72 cm
Meeting mode
On
Door light
100%
Flood light
Off
Temperature
10.2 °C
Motion
Clear
Illuminance
555 lx
Auto-update
Off
OS
Up-to-date
Supervisor
Up-to-date
Core
Up-to-date
DU
https://demo.home-assistant.io/lovelace

Installing Home Assistant



Home Assistant

docker

docker + NixOS

Enable Docker

```
# homeassistant-config.nix
{ config, lib, pkgs, ... }:
{
    virtualisation.docker.enable = true;
    users.extraGroups.docker.members = [ "root" ];
}
```

```
# router-network.nix
...
firewall.extraCommands = ''
    ...
    iptables -A OUTPUT -d 54.227.20.253 -j ACCEPT
    iptables -A OUTPUT -d 54.236.113.205 -j ACCEPT
    iptables -A OUTPUT -d 54.198.86.24 -j ACCEPT
    ...
```

docker + NixOS

Pull Home Assistant

```
docker pull homeassistant/home-assistant
```

```yaml
# compose.yml
services:
  homeassistant:
    container_name: homeassistant
    image: "ghcr.io/home-assistant/home-assistant:stable"
    volumes:
      - /home/root/homeassistant_config:/config
      - /etc/localtime:/etc/localtime:ro
      - /run/dbus:/run/dbus:ro
    restart: unless-stopped
    privileged: true
    network_mode: host
```

docker + VirtualBox

```
VBoxManage modifyvm "vmpsateam03-05" --nat-pf1 "ssh,tcp,,60351,,8123"
```

docker + SSL

```
cd /root/homeassistant_config;
openssl req -sha256 -addext "subjectAltName = IP:192.168.3.5" -newkey rsa:4096 -nodes -keyout pri
# und danach alle Prompts für Country Code, Mail, ... ausfüllen
```

```
...
# add https certificate
http:
  ssl_certificate: /config/fullchain.pem
  ssl_key: /config/privkey.pem
```

Configuring

Home Assistant

Configuring

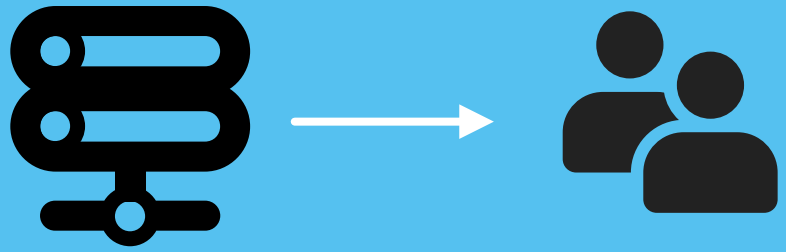
Database

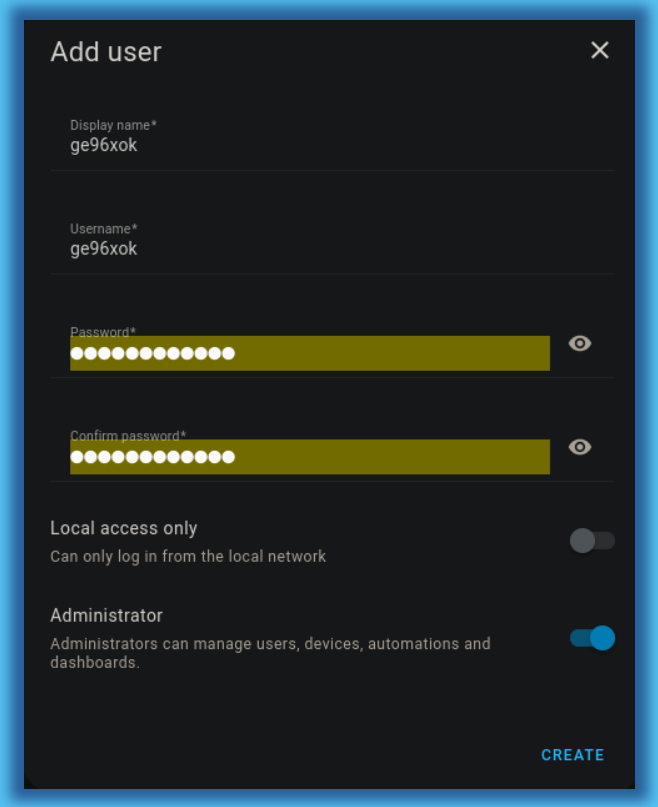
Database

Home Assistant uses databases to store events and parameters for history and tracking. The default database used is SQLite.

```
recorder:
  purge_keep_days: 30
  db_url: mysql://team3:DT7q2K1@192.168.4.5/databaseTeam3
```

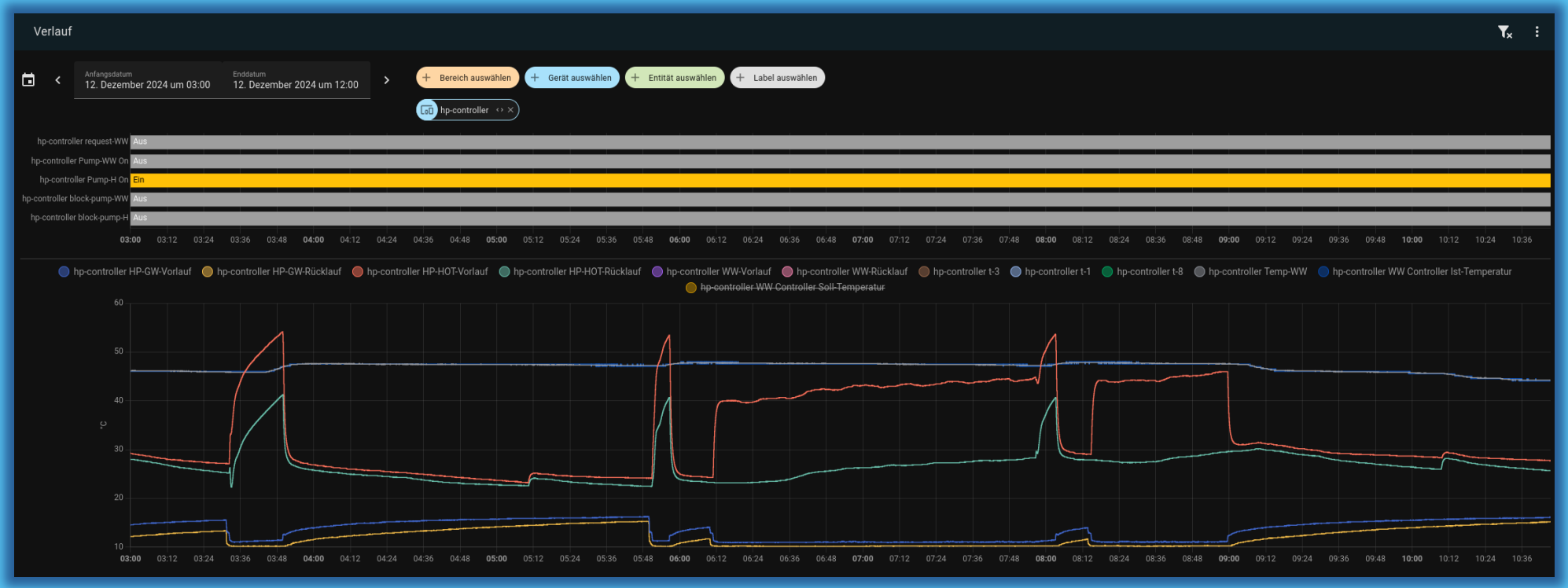
Users




Add user
Display name*
ge96xok
Username*
ge96xok
Password*
Confirm password*
Local access only
Can only log in from the local network
Administrator
Administrators can manage users, devices, automations and dashboards.
CREATE

What Else?

Database + Sensors



Testing

```
# check if docker is installed
if command -v docker 2>&1 >/dev/null; then
    ...
# check if docker config exists
if [ -f /root/compose.yml ]; then
    ...
# check if homeassistant docker container is running
if [ "$( docker container inspect -f '{{.State.Status}}' $container_name )" = "running" ]; then
    ...
# check curl connection
status=$(curl -X GET -o - -I http://131.159.74.56:60351/ | head -n 1)
if [[ $status =~ "200" ]]; then
    ...
```

And now you know

Home Assistant

(and how to install it)